

QualiAdept: Masterclass

QA Manual - From Zero to ISTQB Certified

This syllabus represents the complete structure of the Manual QA course, based on a mix of testing fundamentals, management tools (Jira), technical testing (API & SQL), and real project simulations.

Module 1: Fundamentals and Tester Mindset (Sessions 1-3)

 **Objective:** Understanding the role of QA in the software development process and adopting critical thinking.

Session 1: Basic Concepts and the Psychology of Testing

- What is testing and why is it necessary?
- The difference between Errors, Defects and Failure.
- Risk management (Product vs. project risks).
- **QualiAdept Analogy:** *The tester as a detective vs. The developer as an architect.*

Session 1.2 (Individual Study): The 7 Principles of Testing

- Analysis of ISTQB principles (Exhaustive testing is impossible, Pesticide Paradox, etc.).
- The difference between QA (Quality Assurance) and QC (Quality Control).
- Levels of independence in testing.

Session 2: Testing Planning (Test Plan)

- Process documentation: Strategies, roles and entry/exit criteria.
- Practice: Creating a Testing Plan on the OrangeHRM application.

Session 3: Static Testing

- Reviews and Inspections: Finding defects before the code is written.
- The economic benefits of early testing.

Module 2: Design and Management Techniques (Sessions 4-8)

 **Objective:** Learning methods for writing effective tests and managing them in professional tools.

Session 4: Introduction to Jira & Test Management

- The role of Jira in Agile. Components: Epic, Story, Task, Bug.
- Introduction to TestRail for test case management.

Session 5: Blackbox Techniques (Test Design)

- Equivalence Partitioning (EP) și Boundary Value Analysis (BVA).
- State Transition Testing and Decision Tables.
- **Applied Story:** *How to test an ATM using decision tables.*

Session 6 & 8: Jira & Zephyr Practice

- Fluxul complet: Story -> Test Case -> Cycle Summary -> Execution -> Bug Reporting.
- Generating the Traceability Matrix.
- Monitoring and control reports (Dashboards).

Session 7: Introduction to Whitebox Testing

- Understanding internal structure: Statement Coverage vs. Decision Coverage.
- Exercise: Supermarket loyalty scheme (logical analysis).

Module 3: Web and API Testing (Sessions 9-13)

 **Objective:** Moving from the graphical interface to testing the communication layer (Back-end).

Session 9: GUI Testing (Graphical Interface)

- Checking visual elements (alignment, colors, fonts).
- Using Inspect Element (HTML/DOM) to identify UI defects.

Session 10: API Testing & Postman Basics

- What is an API? The structure of a Request (URL, Method, Headers, Body).
- Response codes (200 OK, 404 Not Found, 500 Server Error).

Session 11 & 12: Advanced Testing in Postman

- Request-uri de tip POST, PUT, DELETE.
- Test automation in Postman (Scripts & Snippets).
- Parsing JSON responses.

Session 13: Running Collections with Newman

- Installing Node.js and Newman.
- Running API tests directly from the Terminal (CLI).
- Integration into CI/CD workflows.

Module 4: Database Testing - SQL (Sessions 14-17)

 **Objective:** Ability to check data integrity directly in the SQL server.

Session 14: Relational Database Theory

- Concepts: Tables, Relationships (1:1, 1:N, N:N), Primary and Secondary Keys.

Session 15: DDL (Data Definition Language) Practice

- Creating the structure: CREATE TABLE, ALTER, TRUNCATE.

Session 16: DML (Data Manipulation Language) Practice

- Populating and modifying data: INSERT, UPDATE, DELETE.

Session 17: DQL (Data Query Language) Practice

- Complex Data query: SELECT, WHERE, JOIN (Inner, Left, Right), GROUP BY, HAVING.

Module 5: Career Preparation and Certification (Sessions 18-19)

 **Objective:** Validation of knowledge through international standards and interview simulations.

Session 18 & 19: ISTQB Foundation Level Preparation

- Exam structure and registration rules.
- Mock Tests Sessions (Exam Simulations).
- Analysis of answers and clarification of trap questions.
- **QualiAdept Tips:** *How to read between the lines on the certification exam.*

Resources and Final Project

- **Final Project:** End-to-end testing of a real application (OrangeHRM/Simple Books API), including Test Plan, Bug Reports in Jira, API Tests in Postman and SQL queries.